

Use Cases and Examples of Prompt Engineering

1. Language and Text Generation

A. Creative Writing

Prompt:

Write a short story about a young woman who discovers a magical portal in her attic.

Output:

One rainy evening, Maya was cleaning the old attic of her grandmother's house when she noticed a small wooden door behind some dusty boxes. She had never seen it before.

When she opened the door, a bright blue light appeared. Curious, Maya stepped through it and found herself in a beautiful world filled with floating islands and glowing flowers.

An old wizard told her that the portal had been waiting for someone brave enough to protect their world. Maya faced several challenges and finally saved the magical kingdom from a dark shadow.

When she returned home, only a few minutes had passed. But Maya knew that her life would never be the same again.

B. Summarization

Prompt:

Summarize the following text: Climate change is causing an increase in global temperatures, melting glaciers, rising sea levels, and extreme weather events. Human activities such as burning fossil fuels and deforestation are major causes. Reducing carbon emissions and using renewable energy can help reduce its impact.

Output:

Climate change is increasing global temperatures and causing problems such as melting glaciers, rising sea levels, and extreme weather. Human activities are the main cause, and renewable energy and reduced carbon emissions can help reduce its effects.

C. Translation

Prompt:

Translate the following text from English to Spanish: "The quick brown fox jumps over the lazy dog."

Output:

El rápido zorro marrón salta sobre el perro perezoso.

D. Dialogue Generation

Prompt:

You are a friendly chatbot helping users troubleshoot their computer problems. Respond to the user's query: "My computer won't turn on."

Output:

I'm sorry you're having trouble with your computer. First, check whether the power cable is properly connected and the power socket is working. If you are using a laptop, make sure the battery is charged. Try holding the power button for 10–15 seconds and then turn it on again. If it still does not start, there may be a hardware or battery issue.

2. Question Answering

A. Open-Ended Question

Prompt:

Explain the concept of quantum computing and its potential impact on the future of technology.

Output:

Quantum computing is a type of computing that uses quantum principles to process information. Unlike classical computers that use bits represented as 0 or 1, quantum computers use qubits, which can represent multiple states.

Quantum computing could solve certain complex problems much faster than traditional computers. It may have applications in cryptography, medicine, artificial intelligence, scientific research, and optimization.

B. Specific Question

Prompt:

What is the capital of France?

Output:

The capital of France is Paris.

C. Multiple Choice Question

Prompt:

Who wrote the Harry Potter series?

- A) J.R.R. Tolkien
- B) J.K. Rowling
- C) Stephen King

Output:

Correct Answer: B) J.K. Rowling

D. Hypothetical Question

Prompt:

What would happen if humans could travel at the speed of light?

Output:

According to current physics, objects with mass cannot reach the exact speed of light because it would require an enormous amount of energy. If humans could somehow travel close to the speed of light, time dilation would occur, meaning time would pass more slowly for the traveler compared to people on Earth.

This could potentially allow humans to travel vast distances in space, although many technological and scientific challenges would need to be solved.

E. Opinion-Based Question

Prompt:

Do you believe that artificial intelligence will eventually surpass human intelligence? Why or why not?

Output:

Artificial intelligence may surpass humans in many specific tasks, such as data analysis, pattern recognition, and complex calculations. However, human intelligence also includes creativity, emotions, social understanding, and ethical judgment.

Therefore, AI may become more powerful in certain areas, but whether it will completely surpass human intelligence in every aspect remains uncertain.

3. Code Generation

A. Python Factorial Program

Prompt:

Write a Python function to calculate the factorial of a given number.

Output:

```
def factorial(n):  
    if n == 0 or n == 1:  
        return 1  
    return n * factorial(n - 1)
```

```
num = 5
print("Factorial of", num, "is", factorial(num))
```

Result:

Factorial of 5 is 120

B. Code Translation

Prompt:

Translate the following Python code to JavaScript:

```
def greet(name):
    print("Hello,", name)
```

Output:

```
function greet(name) {
    console.log("Hello, " + name);
}
```

```
greet("Manikandan");
```

Result:

Hello, Manikandan

C. Code Optimization

Prompt:

Optimize the following Python code to calculate the sum of numbers from 1 to n.

Output:

```
def sum_numbers(n):
    return n * (n + 1) // 2
```

```
n = 100
print("Sum:", sum_numbers(n))
```

Result:

Sum: 5050

D. Code Debugging

Prompt:

Debug the following Java code and explain why it throws a NullPointerException.

Erroneous Code:

```
public class Main {
```

```
public static void main(String[] args) {  
    String name = null;  
    System.out.println(name.length());  
}  
}
```

Explanation:

The error occurs because name has the value null. Calling name.length() attempts to access a method on an object that does not exist.

Corrected Code:

```
public class Main {  
    public static void main(String[] args) {  
        String name = "Manikandan";  
  
        if (name != null) {  
            System.out.println(name.length());  
        }  
    }  
}
```

4. Image Generation

A. Photorealistic Image

Prompt:

A photorealistic image of a beautiful sunset over the ocean, with palm trees silhouetted against an orange and purple sky, cinematic lighting, highly detailed, realistic photography.

Expected Output:

A realistic sunset scene with an ocean, palm trees, and dramatic lighting.

B. Artistic Image

Prompt:

An impressionist painting of a bustling city street with people walking under colorful umbrellas in the rain, inspired by classical impressionist art, visible brush strokes and soft lighting.

Expected Output:

An artistic painting-style image of a rainy city street.

C. Abstract Image

Prompt:

An abstract image representing the concept of hope, using bright colors, flowing shapes, glowing light, and a feeling of optimism.

Expected Output:

A colorful abstract artwork representing hope and positivity.

D. Image Editing

Prompt:

Change the background of this image to a starry night sky and add a bright full moon. Keep the main subject unchanged.

Expected Output:

The original subject remains the same, while the background is changed to a starry night with a full moon.

Conclusion

This practical demonstrated the major applications of Prompt Engineering, including Language and Text Generation, Question Answering, Code Generation, and Image Generation and Editing. Clear, specific, and detailed prompts help AI systems understand the required task and produce more accurate and useful results.